MOTION-04

Class 09 - Science

1. **Assertion (A):** An object can have constant speed but variable velocity. **[1]**
Reason (R): Speed is a scalar but velocity is a vector quantity.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

2. **Assertion (A):** A boy is enjoying a ride on a merry-go-round which is moving at a constant speed of 10 m/s. The boy is in uniform accelerated motion. **[1]**
Reason (R): A body has a uniform acceleration if it travels in a straight line and its velocity first decreases then increases by equal amounts in equal intervals of time.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

3. **Assertion (A):** Motion with uniform velocity is always along a straight-line path. **[1]**
Reason (R): In uniform velocity a motion, speed is the magnitude of the velocity and is equal to the instantaneous velocity.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

4. **Assertion (A):** If a particle is moving with constant velocity, then the average velocity for any time interval is equal to instantaneous velocity. **[1]**
Reason (R): If average velocity of a particle moving on a straight line is zero for a given time interval, then instantaneous velocity at some instant within this interval may be zero.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

5. **Assertion (A):** Motion of satellites around their planets is considered as accelerated motion. **[1]**
Reason (R): During their motion, the speed remains constant, while the direction of motion changes continuously.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

6. **Assertion (A):** An object may acquire acceleration even if it is moving at a constant speed. **[1]**

Reason (R): With change in the direction of motion, an object can acquire acceleration.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

7. **Assertion (A):** When the displacement of a body is directly proportional to the square of the time. Then the body is moving with uniform acceleration. **[1]**

Reason (R): The slope of velocity-time graph with time axis gives acceleration.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

8. **Assertion (A):** The displacement of an object can be either positive, negative or zero. **[1]**

Reason (R): Displacement has both the magnitude and direction.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

9. **Assertion (A):** A boy goes from A to B with a velocity of 20 m/min and comes back from B to A with a velocity of 30 m/min. The average velocity of the boy during the whole journey is zero. **[1]**

Reason (R): The ratio of speed to the magnitude of velocity when the body is moving in one direction is equal to one.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

10. **Assertion (A): T**he speedometer of a car measures the instantaneous speed of the car. **[1]**

Reason (R): Average speed is equal to the total distance covered by an object divided by the total time taken.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

11. **Assertion (A):** A bus moving due north takes a turn and starts moving towards the east with the same speed. There will be no change in the velocity of the bus. **[1]**

Reason (R): Velocity is a vector quantity.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

12. **Assertion (A):** An object may have acceleration even if it is moving with uniform velocity. **[1]**

Reason (B): An object may be moving with uniform velocity but it may be changing its direction of motion.

a) Both A and R are true and R is the correct

b) Both A and R are true but R is not the explanation of A. correct explanation of A.

c) A is true but R is false.
d) A is false but R is true.

13. **Assertion (A):** The speedometer of an automobile measure the average speed of the automobile. **[1]**
Reason (R): Average velocity is equal to total displacement per total time-taken.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

14. **Assertion (A):** The speed of a body can be negative. **[1]**
Reason (R): If the body is moving in the opposite direction of positive motion, then its speed is negative.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

15. **Assertion (A):** Velocity versus time graph of a particle in uniform motion along a straight path is a line parallel to the time axis. **[1]**
Reason (B): In uniform motion the velocity of a particle increases as the square of the time elapsed.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

16. **Assertion (A):** Displacement of a body may be zero when the distance travelled by it is not zero. **[1]**
Reason (R): The displacement is the longest distance between the initial and final position.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

17. **Assertion (A):** The graph between two physical quantities P and Q is a straight line when P/Q is constant. **[1]**
Reason (R): The straight-line graph means that P is proportional to Q or P is equal to constant multiplied by Q.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

18. **Assertion (A):** The speed or velocity of a car running on a crowded city, road changes continuously. **[1]**
Reason (R): The movement of a car on a crowded city road is an example of non-uniform acceleration.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

19. **Assertion (A):** The bus travels 250 km from Delhi to Jaipur towards the West and then comes back to the starting point. Total displacement is zero. **[1]**
Reason (R): The average velocity of the bus for the whole journey (both ways) is 0 kilometers per hour.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false. d) A is false but R is true.

20. **Assertion (A):** A car is said to have a uniform speed of say, 60 km per hour, if it travels 30 km every half hour, 15 km every quarter of an hour, 1 km every minute, and 1/60 km every second. **[1]**

Reason (R): The SI unit of speed is metres per second.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

21. **Assertion (A):** The speed of the car is constant, its velocity is not constant because the direction of the car is changing continuously. **[1]**

Reason (R): The direction of velocity is the same as the direction of displacement of the body.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** A tiger can accelerate from rest at the rate of 4 m/s^2. **[1]**

Reason (R): The velocity attained by it in 10s is 40 m/s.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** The position-time graph of a body moving uniformly in a straight line is parallel to position-axis. **[1]**

Reason (R): The slope of the position-time graph in a uniform motion gives the velocity of an object.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **State True or False:** **[20]**

(i) The velocity of a body is defined as the distance traveled by the body in unit time in a given direction. **[1]**

(ii) When a body moves in a circle, it is called circular motion. **[1]**

(iii) The shortest distance measured from the initial to the final position of an object is known as the displacement. **[1]**

(iv) An object is said to be in uniform motion if it travels equal distances in equal intervals of time, howsoever small the intervals may be. **[1]**

(v) The simplest type of motion is the motion along a straight line called linear motion. **[1]**

(vi) When the velocity of an object changes, we say that the object is accelerating. **[1]**

(vii) The motion of the athlete moving along a circular path is an example of accelerated motion. **[1]**

(viii) An object can be at rest with respect to one thing and in motion with respect to some other thing at the same time. **[1]**

(ix) If the position of an object does not change as time passes, then it is said to be at rest. **[1]**

(x) Negative acceleration is called retardation. **[1]**

(xi) The speed of a body is a vector quantity. **[1]**

(xii) During the uniform motion of an object along a straight line, the velocity remains constant with time. **[1]**

(xiii) The distance-time graph of a body moving with non-uniform speed is a curved line with a variable slope indicating variable speed. **[1]**

(xiv) The velocity of a body is defined as the distance traveled by the body in unit time. **[1]**

(xv) The distance travelled by an object is the length of the actual path traversed by the object during motion. **[1]**

(xvi) When an object travels equal distances in equal intervals of time, it moves with uniform speed. **[1]**

(xvii) The motion of a freely falling body is an example of uniformly accelerated motion. **[1]**

(xviii) Displacement is a scalar quantity. **[1]**

(xix) The distance traveled by an object in motion can never be zero or negative. **[1]**

(xx) The variation in velocity with time for an object moving in a straight line can be represented by a velocity-time graph. **[1]**

25. **Fill in the blanks:** **[17]**

(i) When an object moves in a circular path with uniform speed, its motion is called uniform ________ motion. **[1]**

(ii) If the position of an object changes as time passes, then it is said to be in ________. **[1]**

(iii) Acceleration is a ________ quantity. **[1]**

(iv) An object is said to have ________ motion if it travels unequal distances in equal intervals of time. **[1]**

(v) Negative acceleration is called ________. **[1]**

(vi) Distance is a ________ quantity. **[1]**

(vii) The acceleration is taken to be ________ if it is in the direction of velocity. **[1]**

(viii) A body is said to possess ________ if it travels in a straight line and its velocity increases or decreases by equal amounts in equal intervals of time. **[1]**

(ix) The ________ of an object in motion is the shortest distance between the initial position and the final position of the object. **[1]**

(x) The ________ of a circle of radius r is given by $2\pi r$. **[1]**

(xi) The slope of the velocity-time graph represents the ________ of the body. **[1]**

(xii) The change in velocity of the object for any time interval is ________. **[1]**

(xiii) ________ of a body can be obtained from the slope of the distance-time graph. **[1]**

(xiv) Both distance and displacement have the ________ units. **[1]**

(xv) To locate the position of an object, we have to choose some suitable reference point called the ________. **[1]**

(xvi) ________ is the speed of an object moving in a definite direction. **[1]**

(xvii) ________ of a body is defined as the rate of change of its velocity with time. **[1]**

26. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) Acceleration	(i) Scalar quantity
(b) Distance	(ii) Vector quantity
(c) Speed	(iii) Shortest distance between two positions of the object
(d) Displacement	(iv) The actual path covered by an object

27. Match the following Column A with Column B: **[2]**

Column A	Column B

(a) Acceleration	(i) Distance traveled by the body in unit time
(b) Velocity	(ii) A body covers unequal distances in equal time intervals
(c) Non-uniform speed	(iii) Distance traveled by the body in unit time in a given direction
(d) Speed	(iv) The measure of the change in the velocity of an object per unit time

28. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) The variation in velocity with time for an object moving in a straight line	(i) Equation For Position – Time Relation
(b) The change in the position of an object with time	(ii) Equation For Position – Velocity Relation
(c) $2as = v^2 - u^2$	(iii) Distance – Time Graph
(d) $s = ut + \frac{1}{2} at^2$	(iv) Velocity – Time graph

29. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) Linear motion	(i) The motion repeats itself after regular intervals of time
(b) Circular motion	(ii) The motion along with a straight line
(c) Periodic motion	(iii) The motion which turns or spins about a fixed axis
(d) Rotational motion	(iv) The motion along a circular path

30. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) The distance-time graph for a stationary object	(i) Non-uniform acceleration
(b) The distance-time graph for an object for non-uniform speed	(ii) Straight line to x- axis
(c) The distance-time graph for an object for uniform speed	(iii) Curved line
(d) The velocity changes by unequal amounts in equal intervals of time	(iv) Straight-line parallel to the time axis

31. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) Uniform speed	(i) A graph of distance travelled against time is a curved line
(b) Non-uniform speed	(ii) A graph of distance travelled against time is a straight line
(c) Uniform circular motion	(iii) Changing direction six times to complete one path
(d) Hexagonal path	(iv) When a body moves in a circle

32. Match the following Column A with Column B: **[2]**

Column A	Column B

(a) Speed	(i) The position of an object does not change as time passes
(b) Retardation	(ii) Distance travelled by the body in unit time
(c) An object in motion	(iii) Negative acceleration
(d) An object at rest	(iv) The position of an object changes as time passes

33. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) Velocity	(i) Scalar quantity
(b) Distance	(ii) Vector quantity
(c) Uniform motion	(iii) Simplest type of motion
(d) Linear motion	(iv) Equal distances in equal intervals of time

34. Match the following Column A with Column B: **[2]**

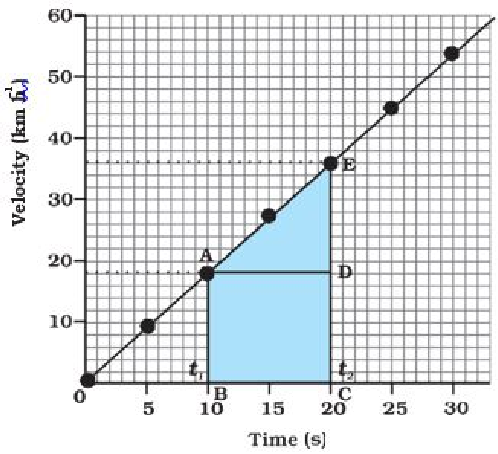
Column A	Column B
(a) Uniform velocity	(i) A body travels in a straight line and its velocity increases or decreases by equal amounts in equal intervals of time
(b) Non-uniform velocity	(ii) A body travels along a straight line, covering equal distances in equal intervals of time
(c) Non-uniform acceleration	(iii) A body covers unequal distances in a particular direction in equal intervals of time
(d) Uniform acceleration	(iv) Velocity changes by unequal amounts in equal intervals of time

35. Match the following Column A with Column B: **[2]**

Column A	Column B
(a) Acceleration	(i) Negative acceleration
(b) Speed	(ii) A body travels in a straight line and its velocity increases or decreases by equal amounts in equal intervals of time
(c) Retardation	(iii) Measure of the change in the velocity of an object per unit time
(d) Uniform acceleration	(iv) Distance traveled by the body in unit time

36. **Read the following and answer any four questions:** **[4]**

In the velocity-time graph for the motion of the car. The nature of the graph shows that velocity changes by equal amounts in equal intervals of time. For all uniformly accelerated motion, the velocity-time graph is a straight line.



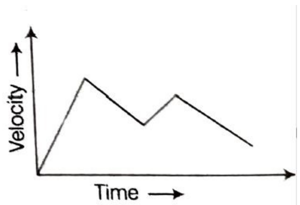
i. The slope of a velocity-time graph gives

a. the distance
b. the displacement
c. the acceleration
d. the speed

ii. Which of the following statement is correct regarding the velocity and speed of a moving body?

a. The velocity of a moving body is always higher than its speed.
b. The speed of a moving body is always higher than its velocity.
c. The speed of a moving body is its velocity in a given direction.
d. The velocity of a moving body is its speed in a given direction.

iii. The following graphs shows:



a. uniformly accelerated motion
b. non-uniformly accelerated motion
c. non- uniformly decelerated motion
d. uniformly decelerated motion

iv. If the displacement of an object is proportional to square of time, then the object moves with

a. uniform velocity
b. uniform acceleration
c. increasing acceleration
d. decreasing acceleration

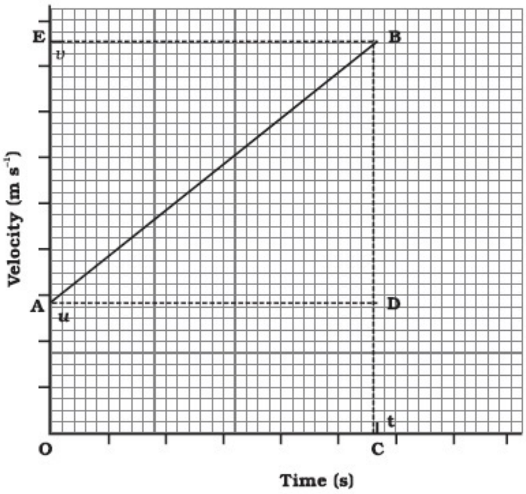
v. which of the following is an incorrect match:

	column I	column II
a.	Straight-line parallel to time axis in the velocity-time graph	Scalar quantity
b.	The slope of the velocity-time graph gives	Shortest distance between the initial and final position
c.	Uniform circular motion	Rate of change of velocity with respect to time

d.	Displacement	Body in uniform motion

37. **Read the following and answer any four questions:** **[4]**

In the velocity-time graph of an object that moves under uniform acceleration as the initial velocity of the object is u (at point A) and then it increases to v (at point B) in time t. The velocity changes at a uniform rate a.



i. A boy goes from A to B with a velocity of 20 m/min and comes back from B to A with a velocity of 30 m/min. The average velocity of the boy during the whole journey is

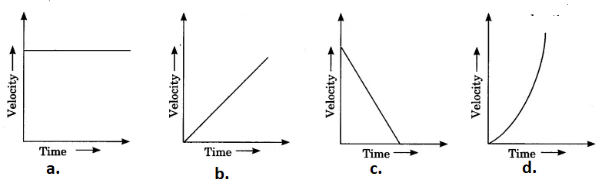
a. 24 m/min

b. 25 m/s

c. Zero

d. 20 m/min

ii. A car is moving along a straight road with uniform velocity. It is shown in the graph.



iii. The ratio of speed to the magnitude of velocity when the body is moving in one direction is

a. less than one

b. greater than one

c. equal to one

d. greater than or equal to one

iv. Which of the following statement is correct with respect to the velocity-time graph given above?

I. the perpendicular lines BC and BE are drawn from point B on the time

II. initial velocity is represented by OA

III. the final velocity is represented by OC

IV. the time interval t is represented by OB.

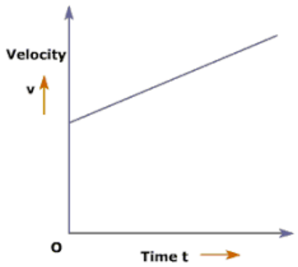
a. (I) and (II)

b. (II) and (III)

c. (IV) and (I)

d. (III) and (IV)

v. According to the given velocity-time graph, the object



a. is moving with uniform velocity

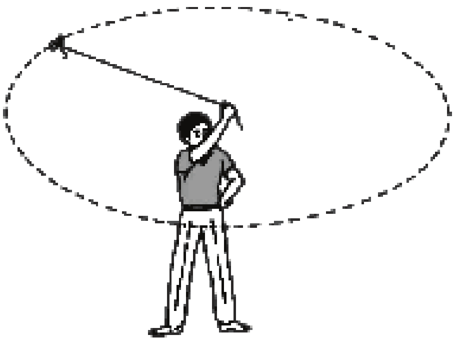
b. has some initial velocity

c. is moving uniformly with some initial velocity

d. is at rest

38. **Read the following and answer any four questions:** **[4]**

When an object moves in a circular path with uniform speed, its motion is called uniform circular motion. The direction of motion changed at every point moving along the circular path.



i. Which one of the following is most likely not a case of uniform circular motion?

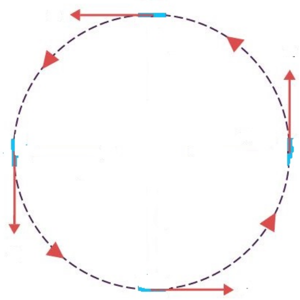
a. The motion of the earth around the sun.

b. The motion of a toy train on a circular track.

c. The motion of a racing car on a circular track.

d. The motion of hours' hand on the dial of a clock.

ii. The train is moving on a track**(below image)**. Though the speed of a train is constant the direction of motion (or direction of speed) is changing continuously. So, the train is exhibiting:



a. uniform notion

b. decelerated motion

c. uniform motion

d. accelerated motion

iii. A cyclist goes around a circular track once every 2 minutes. If the radius of the circular track is 105 metres, calculate his speed.

a. 5.5 m/s

b. 5.6 m/s

c. 5.7 m/s

d. 5.8 m/s

iv. Which of the following statement is correct?

I. Motion of the moon and the earth is an example of non-uniform circular motion.

II. When the velocity of an object changes, we say that the object is accelerating.

III. A satellite in a straight orbit around the earth.

IV. The change in the velocity could be due to a change in its magnitude or the direction of the motion or both.

a. (I) and (II)

b. (II) and (III)

c. (III) and (IV)

d. (II) and (IV)

v. A sprinter is running along the circumference of a big sports stadium with constant speed. Which of the following do you think is changing in this case?

a. The magnitude of acceleration is produced.

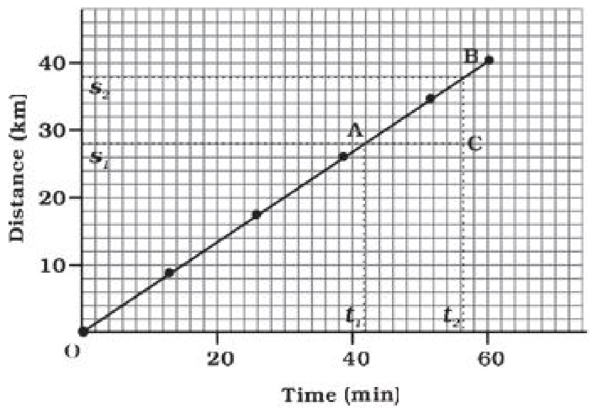
b. Distance covered by the sprinter per second.

c. The direction in which the sprinter is running.

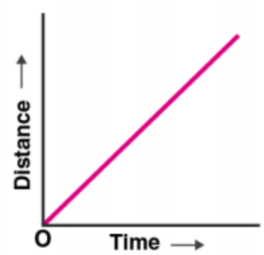
d. The centripetal force acting on the sprinter.

39. **Read the passage and answer any four questions:** **[4]**

Graphical representation of the distance-time graph of moving body at a uniform speed. when an object travels equal distances in equal intervals of time, it moves with uniform speed.



i. What conclusion can you draw about the speed of a body from the following distance-time graph?



a. Uniform speed

b. Non-uniform speed

c. Uniform velocity

d. Non-uniform velocity

ii. Which of the following statement is incorrect about the graphical representation of motion?

i. A straight line graph helps in solving a linear equation
ii. Line graphs show the dependence of one physical quantity
iii. In the distance-time graph, time is taken along the y–axis
iv. In the distance-time graph, distance is taken along the x-axis

a. (I) and (II)
b. (II) and (III)
c. (II) and (IV)
d. (III) and (IV)

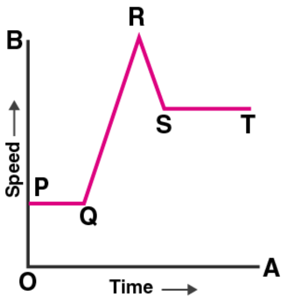
iii. The area under a speed-time graph represents a physical quantity which has the unit of :

a. m
b. m^2
c. ms^{-1}
d. ms^{-2}

iv. A bus moving along a straight line at 20m/s undergoes an acceleration of 4 m/s^2. After 2 seconds, its speed will be :

1. 8 m/s
2. 12 m/s
3. 16 m/s
4. 28 m/s

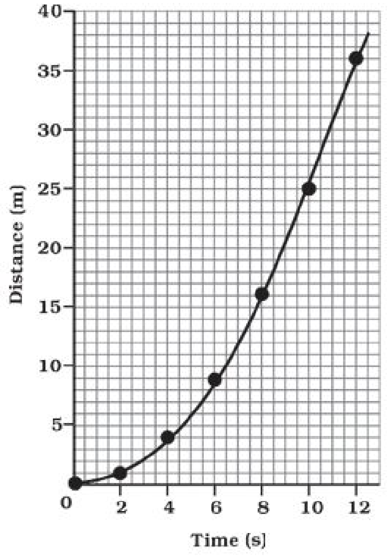
v. A student draws a distance-time graph for a moving scooter and finds that a section of the graph is a horizontal line parallel to the time axis. Which of the following conclusion is correct about this section of the graph?



a. The scooter has uniform speed in this section
b. The distance travelled by scooter is the maximum in this section
c. The distance travelled by scooter is the minimum in this section
d. The distance travelled by scooter is zero in this section

40. **Read the passage and answer any four questions:** **[4]**

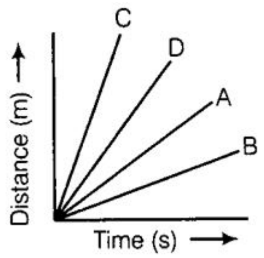
The change in the position of an object with time can be represented on the distance-time graph adopting a convenient scale of choice. In the distance-time graph, time is taken along the x–axis and distance is taken along the y-axis.



i. A man travels a distance of 1.5 m towards East, then 2.0 m towards South and finally 4.5 m towards East. What is the total distance traveled?

a. 8m
b. 16m
c. 5m
d. 7m

ii. Four cars A, B, C and D are moving on a levelled road. Their distance versus time graphs are shown in the adjacent figure. Choose the correct statement.



a. Car A is faster than car D
b. Car B is the slowest
c. Car D is faster than car C
d. Car C is the slowest

iii. If the displacement of an object is proportional to the square of time, then the object is moving with:

a. uniform velocity
b. uniform acceleration
c. increasing acceleration
d. decreasing acceleration

iv. Which of the following statement is correct for distance-time graph?

I. Time is taken along the x–axis
II. When an object travels equal distances in equal intervals of time, it moves with uniform speed
III. Distance-time graphs can be employed under various conditions
IV. For uniform speed, a graph of distance travelled against time is a curved line

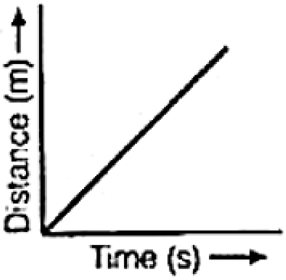
a. (I), (II) and (III)

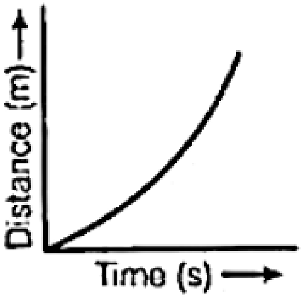
b. (II) and (IV)

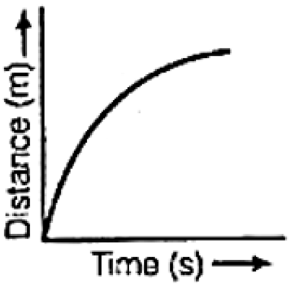
c. (III) and (II)

d. (I) and (III)

v. Which of the following figures correctly represents uniform motion of a moving object?

a. 

b. 

c. 

d.